SSLV, PSLV and LVM3

Three launch vehicles, one toolkit: every idea from Chapters 0 to 6 applied to ISRO’s small, medium and heavy launchers.

Isagi · November 2026 · Theory from NASA Glenn’s Beginner’s Guide to Rockets; LVM3 figures from ISRO, checked 9 November 2026; SSLV and PSLV figures from public references

HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- This is a review chapter. Before each example, say which earlier chapter’s idea it uses, then do the sum.
- LVM3 data are ISRO’s nominal values (Chapter 0). SSLV and PSLV data are from public references that disagree with themselves in places (listed in Sources). Treat comparisons as good to a few percent, not exact. Use $g = 9.81\ m/s^2$.

By the end of this chapter you can: compare three launchers by propellant share, payload fraction, lift-off thrust-to-weight and average specific impulse; explain from first principles why each is built the way it is; and choose a vehicle for a given job.

1 Three vehicles, three jobs

	SSLV	PSLV-XL	LVM3
Height	34 m	44 m	43.5 m
Diameter	2 m	2.8 m	4 m (5 m fairing)
Lift-off mass	120 t	320 t	640 t
Stages	3 solid + liquid trimming module	solid, liquid, solid, liquid (+ 6 solid strap-ons)	2 solid boosters, liquid core, cryogenic upper
Payload	about 500 kg to low Earth orbit (500 km)	3,800 kg LEO (200 km); 1,750 kg SSO (620 km); 1,300 kg GTO	8,000 kg LEO; 4,000 kg GTO
Source	public	public	ISRO

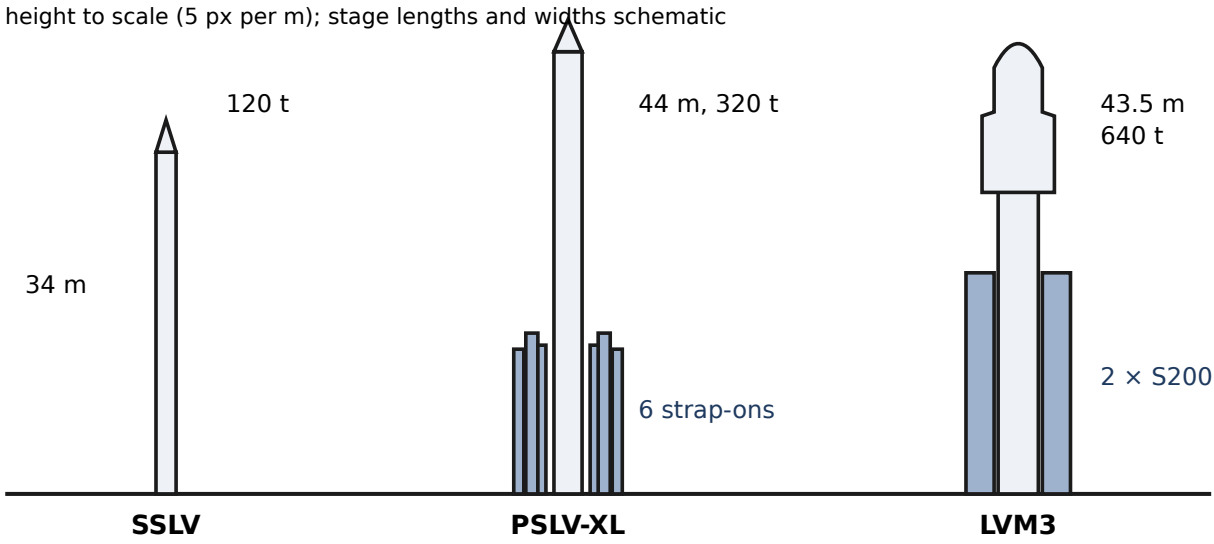


Figure 1. The three vehicles at one height scale. Darker shapes are strap-on solid boosters. PSLV and LVM3 are almost the same height; LVM3 is twice as heavy and wider.

ISRO LENS: A LADDER OF JOBS

The names tell the story. SSLV (Small Satellite Launch Vehicle) serves hundreds-of-kilogram payloads on demand. PSLV (Polar Satellite Launch Vehicle) is the workhorse for 1 to 2 t into sun-synchronous orbit. LVM3 (Launch Vehicle Mark 3) takes 4 t to geosynchronous transfer orbit. Each step is two to three times heavier than the last.

2 Where the propellant sits (Chapters 0, 1, 2)

Every launcher is mostly propellant, and the type of propellant sets the efficiency. Chapter 1 gave the rule: solids give about 260 to 275 s, storable liquids about 290 to 295 s, and LOX/LH_2 about 443 s. The bars below show how many tonnes of each type each vehicle carries.

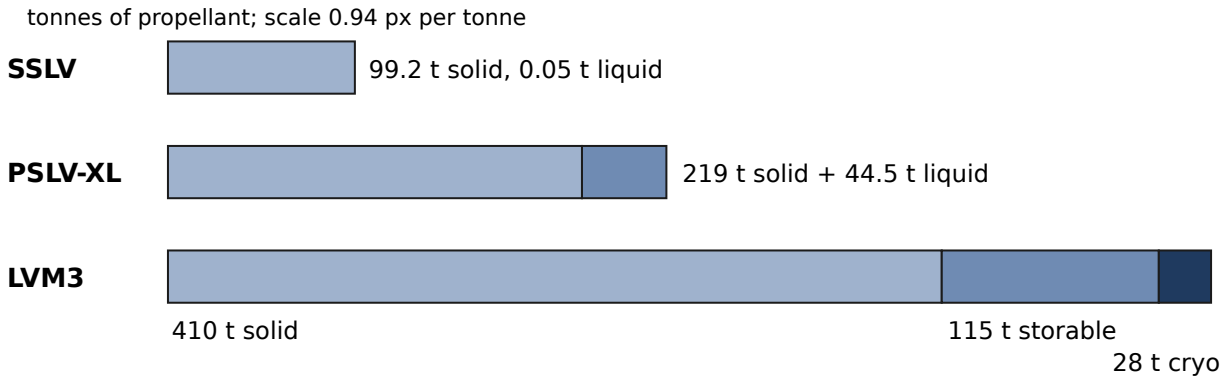


Figure 2. Propellant by type. Light blue: solid. Mid blue: storable liquid. Dark blue: cryogenic. Only LVM3 carries any cryogenic propellant, and it is just 5% of the total.

3 How well does each rocket deliver? (Chapters 0, 2)

The payload fraction (payload divided by lift-off mass) is the simplest efficiency score. Because orbits differ (200 km at 30° versus 500 km versus GTO), compare within a column, and treat the numbers as rough.

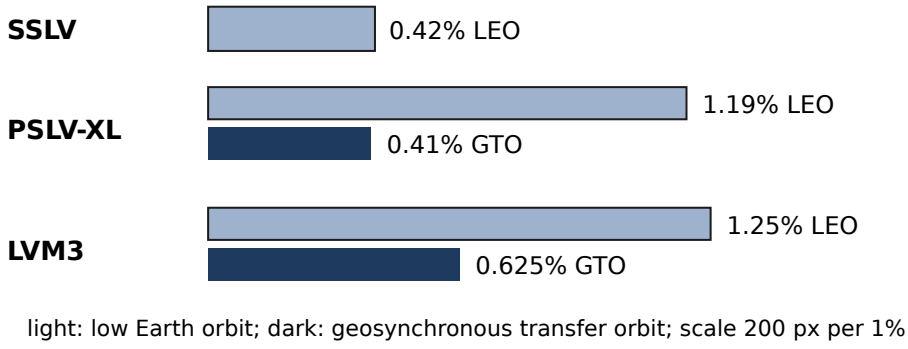


Figure 3. Payload as a share of lift-off mass. LVM3's extra efficiency shows up most for GTO.

4 Worked examples

Example A: propellant ledger (Chapter 0)

```
SSLV: 87,000 + 7,700 + 4,500 + 50 = 99,250 kg
      99,250 / 120,000 = 82.7%
PSLV: 138,200 + 6 × 12,200 + 42,000 + 7,600 + 2,500
      = 263,500 kg;  263,500 / 320,000 = 82.3%
LVM3: 2 × 205,000 + 115,000 + 28,000 = 553,000 kg
      553 / 640 = 86.4%
```

All three are over 80% propellant. LVM3's share is higher, which fits the general pattern that larger vehicles carry proportionally less structure.

Example B: payload fraction (Chapter 2)

```
SSLV  LEO: 500 / 120,000   = 0.42%
PSLV  LEO: 3,800 / 320,000 = 1.19%
      SSO: 1,750 / 320,000 = 0.55%
      GTO: 1,300 / 320,000 = 0.41%
LVM3  LEO: 8,000 / 640,000 = 1.25%
      GTO: 4,000 / 640,000 = 0.625%
GTO ratio: 0.625 / 0.406 ≈ 1.5
```

For the same trip to GTO, LVM3 delivers about 1.5 times as much payload per tonne of lift-off as PSLV. That gain is Chapter 1's cryogenic upper stage at work.

Example C: lift-off thrust-to-weight (Chapters 0 and 3)

Use peak thrust from public references (an upper bound; Chapter 1 showed average thrust is lower).

```
SSLV: weight 120,000 × 9.81 = 1,177 kN
      peak thrust 2,496 kN       → T/W ≈ 2.12
PSLV: weight 320,000 × 9.81 = 3,139 kN
      4,847 + 6 × 703.5 = 9,068 kN → T/W ≈ 2.89
LVM3: weight 640,000 × 9.81 = 6,278 kN
      2 × 5,150 = 10,300 kN      → T/W ≈ 1.64
```

The PSLV figure assumes all six strap-ons burn together; if fewer are lit at lift-off, its T/W is lower (Practice problem P4). Even so, the pattern holds: the small rocket leaves the pad briskly, and heavy LVM3 lifts off more gently. A smaller T/W means a longer climb and a larger gravity loss (Chapter 3), the price of a bigger payload.

Example D: average specific impulse (Chapters 1 and 6)

Weight each stage's I_{sp} by the propellant it carries.

```
LVM3: (410×274.5 + 115×293 + 28×443) / 553
    = (112,545 + 33,695 + 12,404) / 553
    = 158,644 / 553 ≈ 287 s
PSLV: (138.2×269 + 73.2×262 + 42×293
       + 7.6×295 + 2.5×308) / 263.5
    = 71,672 / 263.5 ≈ 272 s
```

LVM3's cryogenic stage has I_{sp} of 443 s but carries only 5% of the propellant, so it lifts the vehicle average by just 15 s over PSLV's. Efficiency is won at the top of the stack, where each kilogram is worth the most (Chapter 2), not by raising the average.

Example E: choose the vehicle (all chapters)

```
Job 1: a 600 kg satellite to a 500 km orbit
   SSLV: about 500 kg (public figure), short by 100 kg.
   Choose PSLV.
Job 2: a 3,000 kg communication satellite to GTO
   PSLV: 1,300 kg.  LVM3: 4,000 kg.  Only LVM3 can do it.
Job 3: a 150 kg demonstrator, on demand
   SSLV fits; on PSLV it would use under 5% of capacity.
```

Matching payload to the smallest vehicle that fits is the logic of the whole ladder. SSLV's final liquid velocity-trimming module (50 kg of propellant, small thrusters) fine-tunes the last bit of speed, the same idea as PSLV's final liquid stage.

CHAPTER SUMMARY

- SSLV 120 t, PSLV-XL 320 t, LVM3 640 t: each step is two to three times heavier than the last.
- All three are 82% to 86% propellant. Only LVM3 carries cryogenic propellant, and just 5% of its total.
- Payload fraction to LEO: 0.42% (SSLV), 1.19% (PSLV), 1.25% (LVM3). To GTO: LVM3 beats PSLV by about 1.5 times.
- Peak lift-off T/W falls with size: about 2.1 (SSLV), at most 2.9 (PSLV), 1.6 (LVM3).
- Average I_{sp} is about 287 s for LVM3 and 272 s for PSLV. Top-stage efficiency matters most.

Series complete. You can now read any launch-vehicle spec sheet and say what each number means.

Practice problems

Use g = 9.81 m/s². Answers are on the last page.

P1. What share of SSLV's propellant is in its three solid stages? What share of its lift-off mass is the liquid trimming module's 50 kg?

P2. Find PSLV-XL's payload fraction to GTO and to the sub-GTO orbit (1,425 kg).

P3. How many kilograms of payload to GTO per tonne of lift-off mass does each of PSLV-XL and LVM3 deliver?

P4. If only 4 of PSLV-XL's 6 strap-ons were burning at lift-off (with the core), what would peak T/W be?

P5. What share of PSLV-XL's propellant is solid? How does that compare with LVM3's solid share?

P6. You need to launch 1,000 kg into a 620 km sun-synchronous orbit. Which of the three vehicles would you choose, and why not the other two?

Glossary

SSLV
Small Satellite Launch Vehicle; three solid stages and a liquid trimming module.

PSLV
Polar Satellite Launch Vehicle; four stages alternating solid and liquid, with strap-ons in the XL version.

LVM3
Launch Vehicle Mark 3; solid boosters, liquid core and cryogenic upper stage.

Strap-on
A booster fixed alongside the core stage.

Velocity trimming module (VTM)
SSLV's small liquid stage that fine-tunes the final speed.

LEO, SSO, GTO
Low Earth orbit; sun-synchronous orbit; geosynchronous transfer orbit.

Payload fraction
Payload mass divided by lift-off mass.

Propellant-weighted I_{sp}
Average I_{sp} with each stage weighted by its propellant mass.

Peak T/W
Maximum thrust divided by lift-off weight; an upper bound on real lift-off acceleration.

Answer key

Fold this page back before you start the problems.

P1. Solids: 87,000 + 7,700 + 4,500 = 99,200 kg out of 99,250 kg, which is 99.95%. VTM: 50 / 120,000 = 0.04% of lift-off mass.

P2. GTO: 1,300 / 320,000 = 0.406%. Sub-GTO: 1,425 / 320,000 = 0.445%.

P3. PSLV: 1,300 / 320 = 4.06 kg/t. LVM3: 4,000 / 640 = 6.25 kg/t.

P4. Thrust = 4,847 + 4 × 703.5 = 7,661 kN. T/W = 7,661 / 3,139 ≈ 2.44.

P5. PSLV solids: 138,200 + 73,200 + 7,600 = 219,000 kg out of 263,500 = 83.1%. LVM3 solids: 410 / 553 = 74.1%. PSLV is more solid-heavy; LVM3 trades some solid for liquid and cryogenic propellant.

P6. PSLV-XL (1,750 kg to SSO at 620 km). SSLV carries only about 300 kg to 500 km SSO, far short. LVM3 could carry it, but it is twice as heavy and would waste most of its capacity.

Sources

Theory: NASA Glenn Research Center, Beginner’s Guide to Rockets. LVM3: ISRO’s LVM3 launch vehicle page (lift-off mass 640 t, propellant masses, 8 t LEO and 4 t GTO). SSLV and PSLV: public references (Wikipedia entries, retrieved 10 November 2026), which disagree with themselves in places. SSLV: LEO payload 500 kg in the text, 375 to 525 kg in the infobox; no Isp given. PSLV: stage thrust, propellant and burn times differ between the infobox and text (for example PS2 propellant 42 t versus 41.5 t, burn 151 s versus about 130 s); the SSO payload is 1,750 or 1,800 kg. Stage I_{sp} values for PSLV and the S200 are from the same references. The comparisons here are good to a few percent.